

# The Role of Communication in Agile Systems Development

## An Analysis of the State of the Art

Agile systems development (SD) promotes a shift from the traditional, document-driven communication paradigm towards more informal, face-to-face communication. Our current state of knowledge about the implementation of this changing communication paradigm in agile SD projects is fragmented because of the lack of an integrative lens across different studies. We provide an integrated view of the role of communication in agile SD by conducting a structured, systematic literature review. We find that despite its acknowledged importance, our knowledge of communication and agile SD is limited because hardly any studies open up the communication process. In consequence, we present several research gaps.

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## 1 Introduction

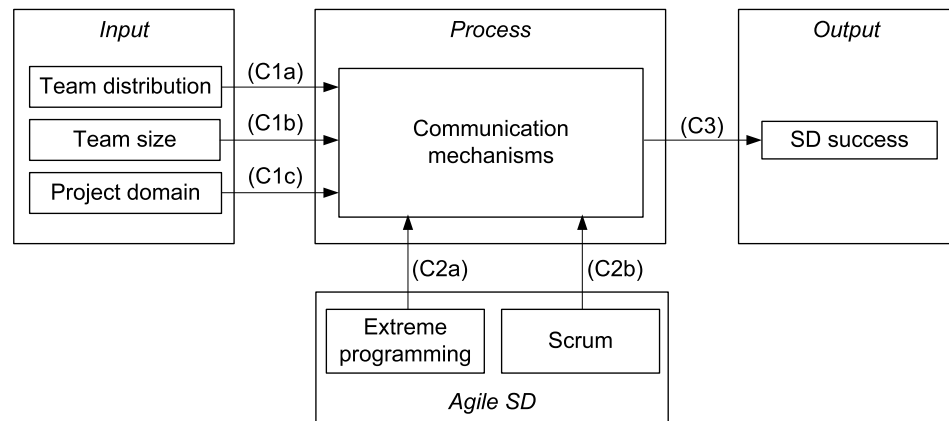
Agile development methods such as Extreme Programming (XP) (Beck 1999) or Scrum (Schwaber and Beedle 2002) are very popular in today's software industry. Agility is the continual readiness of a systems development (SD) method "to rapidly or inherently create change, proactively or reactively embrace change, and learn from change while contributing to perceived customer value" (Conboy 2009, p. 340). Those methods complement iterative and prototyping approaches to SD and have been suggested as a way to react quickly to changing requirements by emphasizing small release cycles and continuous integration of the customer (Erickson et al. 2005). Agile SD methods appear to incorporate most of the practical lessons learned about SD during the past: they trade strict control for more flexibility and autonomy, the overall development pro-

cess is not planned and scheduled upfront, and progress is made in small iterative phases while encouraging change and constant feedback (Cao et al. 2009; Vidgen and Wang 2009).

Even though numerous concepts as well as research frameworks have been proposed and several reviews have highlighted the considerable amount of past research efforts, research lags behind practice in understanding agile SD (Conboy 2009; Dingsøyr et al. 2012; Dybå and Dingsøyr 2008; Erickson et al. 2005). In sum, there is "a backlog of research problems to be solved" (Rajlich 2006, p. 70). One of the most pressing issues is the missing "theoretical glue" (Conboy 2009, p. 330) of agile SD. The principles and practices of agile SD have mainly been derived from past experiences, and the agile SD literature is lacking theoretical foundation (Lee and Xia 2010), which has led to calls for a more theory-based approach in research on agile SD (Dingsøyr et al. 2012).

One of the central aspects of agile SD methods that differentiates them from traditional, plan-driven SD methods is the central role of communication, both within the development teams as well as with customers (Melnik and Maurer 2004; Pikkariainen et al. 2008; Sarker et al. 2009). The Agile Manifesto (Beck et al. 2001), which formulates common values and principles of agile SD, highlights the fundamental role of informal communication for agile SD: business customers

**Fig. 1** Theoretical framework and selected categories based on the Unified Model of SD Success (adapted from Siau et al. 2010)



and developers should work together continuously and daily, and project information is to be shared by means of informal communication such as face-to-face conversation rather than through formal documentation, plans, and models. In consequence, agile SD is also described as a “cooperative game of invention and communication” (Cockburn 2002, p. 28), which underlines the importance of human and social factors (Dybå and Dingsøyr 2009). Some agile approaches rely “almost completely on oral communication with the customer” (Ramesh et al. 2010, p. 451).

Even though communication is repeatedly stated to be one of the fundamental aspects of agile SD (Melnik and Maurer 2004; Pikkarainen et al. 2008; Sarker et al. 2009) and “the importance of communication has been shown paramount in agile development” (Korkala et al. 2006, p. 76), few studies have attempted to understand and explain the interplay between agile SD and communication (Maruping et al. 2009a; Pikkarainen et al. 2008). We therefore provide an integrative lens on research studies concerning agile SD and communication by conducting a systematic literature review, thereby addressing the following research question: “What is the role of communication in SD projects employing agile practices?” The goals of this review are (1) to report on past results, (2) to identify and structure the gaps in current knowledge on the role of communication in agile SD, and (3) to provide guidance for future research.

The remainder of this paper is structured as follows. In the next section, we present our literature review design, including the source selection and the theoretical basis used for framing the search.

Next, we present the results of the review and analyze in detail how prior research has addressed communication in agile SD. Subsequently, we summarize our findings and give implications for future research by highlighting several research gaps.

## 2 Review Design

We conduct a structured, systematic literature review in the field of agile SD in order to investigate the extent to which communication has been investigated so far. Following the guidelines of Webster and Watson (2002), we specifically focus on the leading journals and conferences of the relevant disciplines for agile SD, which is extended by a database search (Sect. 2.1). We use a concept matrix that is based on several categories to structure the presentation of the results and to summarize major findings and insights. In order to derive the categories for our literature review, we develop a theoretical framework that is based on the Unified Model of SD Success (Siau et al. 2010) (Sect. 2.2).

### 2.1 Data Collection and Search Process

We split the search process into two parts (cf. Online-Appendix A for a more detailed description of the search process). In the first stage, we selected all relevant papers within the most popular journals and conferences on agile SD in the software engineering domain (according to Dingsøyr et al. 2012). Since agile SD is also relevant for the information systems (IS) domain, we included the Senior Scholar’s Basket of Journals<sup>1</sup> as well as the

ICIS conference as the premier IS conference. We scanned each paper manually for communication-related findings. Additionally, we used the automatic search function in order to search each paper for the term “communication”.

In the second stage of the search process, we extended our initial results by searching within eight databases for the term “communication” used in the title, abstract, and keywords of papers on agile ISD. During both search stages we scrutinized the titles and abstracts of the papers and excluded papers that did not focus on agile SD. For example, many papers focused on ‘organizational agility’ or ‘supply chain agility’.

This search process resulted in 333 papers in total that included relevant empirical findings on agile SD in combination with communication. The findings of the 333 papers on communication in relation to agile SD were copied into a separate file, which provided the basis for the structuring of the data.

### 2.2 Theoretical Framework and Categories

We used the Unified Model of SD Success (Siau et al. 2010) to derive the categories for structuring and analyzing the results of the identified papers. Based on a comprehensive literature review, the model synthesizes past research efforts on SD and identifies important factors that influence the SD process. This model provides a suitable framework for our investigation because it gives a summary representation of the SD process and its input as well as output factors (cf. Fig. 1).

Category 1 (C1) is concerned with the input dimension of the framework. Agile SD is considered to be mainly

<sup>1</sup><http://home.aisnet.org/displaycommon.cfm?an=1&subarticlenbr=346>.

**Table 1** Analytic categories (concept matrix)

| Category                      | Short Description                                                                                                                                                                                                                    |
|-------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Category 1<br>(C1a, C1b, C1c) | Influence of team distribution on communication mechanisms of agile SD teams.<br>Influence of team size on communication mechanisms of agile SD teams.<br>Influence of project domain on communication mechanisms of agile SD teams. |
| Category 2<br>(C2a, C2b)      | Influence of agile methods and practices on communication mechanisms of agile SD teams.                                                                                                                                              |
| Category 3 (C3)               | Influence of communication mechanisms of agile SD teams on SD success.                                                                                                                                                               |

sued for small, co-located teams that develop non-critical software (Ågerfalk et al. 2009). In order to investigate this claim, we include team distribution (Category C1a), team size (Category C1b), and project domain (Category C1c) as separate categories in our framework because these input variables presumably should influence the communication mechanisms of SD teams (Ågerfalk et al. 2009):

- Team distribution refers to the extent to which team members of SD projects are separated from each other (distributed) or close to each other (co-located) in terms of the physical location (Ågerfalk et al. 2009).
- Team size is defined as the number of team members on the SD project team (e.g., Ågerfalk et al. 2009; Maruping et al. 2009a).
- The project domain of the SD project deals with project characteristics such as whether a software or application system is developed for a single (in-house) customer or for commercial release, whether it is developed from scratch or an existing system is enhanced, and whether it is non-critical or mission-critical (Ågerfalk et al. 2009).

The SD process is also influenced by the SD method employed. In our case, the methods are approaches for agile SD, which can be drilled down to specific agile practices in use. We specifically examine how the most widely used agile methods in industry (Dingsøyr et al. 2012), namely Extreme Programming (XP) (Category C2a) and Scrum (Category C2b), influence the communication mechanisms of SD teams. Both methods promote several techniques and practices; for example XP includes technical practices such as pair programming,

continuous integration, and refactoring (Beck 1999), whereas Scrum is focused on project management practices such as daily stand-ups, sprint planning, and review meetings (Schwaber and Beedle 2002).

The outcome of the process dimension is SD success, which consists of factors such as project success, system usage, and user satisfaction (Siau et al. 2010). We examine how the communication mechanisms of agile SD teams influence SD success (Category C3).

The selected categories are summarized in Table 1.

### 3 Analysis and Results

#### 3.1 Influence of Input Factors on Communication Mechanisms

##### 3.1.1 Team Distribution (Category 1a)

155 studies address topics and issues related to team distribution and communication. Teams in co-located contexts are reported to communicate more efficiently due to physical proximity, for example, by using co-located office spaces, which enable collaboration by whiteboards, status boards, and other informal communication media (e.g., Fruhling and de Vreede 2006; Mishra et al. 2012). An important factor for co-located teams is the layout of the working space (Melo et al. 2011), for example, half height glass barriers and half cubicles are reported to improve the frequency and quality of communication compared to full cubicles (Mishra et al. 2012). However, some studies have found that co-located, open office spaces can also hinder communication because it is hard to focus and concentrate due to the high noise level

(Layman et al. 2004; Pikkarainen et al. 2008). In contrast, others find that conversations in co-located office spaces are not perceived as distracting (Whitworth and Biddle 2007).

In distributed contexts, the “lessons of extreme distributed development boiled down ... to one thing – Communications” (Poole 2004, p. 67). The critical role of communication is also confirmed by many other studies (e.g., Ramesh et al. 2012; Sarker et al. 2009; Sarker and Sarker 2009). However, other investigations indicate that communication is not the key factor; they consider a well-defined customer to be most important for distributed projects (Korkala and Abrahamsson 2007).

Challenges in distributed contexts that are connected to communication are reported to stem mainly from different time zones, languages, and cultures (Dorairaj et al. 2012b; Ramesh et al. 2012; Sarker and Sarker 2009). In consequence, the amount of verbal communication is lower in distributed projects compared to co-located teams (Niinimäki 2011), which may result in miscommunication, misunderstandings, and confusion among team members (Hossain et al. 2009b).

Several studies claim that agile methods help to solve this major communication challenge of distributed projects by forcing team members to communicate, thus reducing the temporal, geographic, and socio-cultural distances in distributed projects (Holmström et al. 2006; Paasivaara et al. 2008). Communication challenges of distributed projects may also be mitigated by distributing work within the same time zone as well as by synchronizing work hours, which may entail working beyond normal office hours (Hossain et al. 2009a; Paasivaara et al. 2008; Sindhgatta et al. 2011). Trusted relationships as well as a shared understanding are also reported as crucial (Holmström et al. 2006; Lavazza et al. 2010). For example, trust can be built by “seeding visits” to other sites at the beginning of the project as well as by “maintaining visits” in later stages (Boden et al. 2007; Paasivaara et al. 2008). It is recommended that a key member of one team should be physically located with the other team in order to facilitate communication and bridge technical, distance, and language barriers (e.g., Braithwaite and Joyce 2005; Dorairaj et al. 2012b; Layman et al. 2006).

Research shows that distributed agile teams rely heavily on communication tools in order to collaborate across several time zones (Bocock and Martin 2011; Dorairaj et al. 2012b). This leads to the recommendation that multiple communication modes, including a variety of differing media, should be provided in order to bridge breakdowns of communication tools and to comply with differing preferences (Paasivaara et al. 2008, 2009).

Asynchronous media for sharing information and knowledge such as e-mail (Korkala et al. 2009), wikis, ticketing systems, and a shared codebase have been shown to help to reduce the need for synchronous communication (Braithwaite and Joyce 2005; Hossain et al. 2009a; Reeves and Zhu 2004). In terms of synchronous communication media, telephone and video conferencing (also combined with desktop sharing) have been found more effective for maintaining constant communication than asynchronous media (e.g., Dorairaj et al. 2012b; Layman et al. 2006; Young and Terashima 2008). Videoconferencing should be preferred when direct face-to-face communication is not possible because it most resembles natural communication (Bless 2010). Instant messaging is another frequently employed medium because it is deemed to be a good compromise between synchronous and asynchronous communication (Estler et al. 2012). In sum, it seems to be important to choose the right communication media for the right task, for example, synchronous media is suitable for solving queries, but e-mail remains suitable for stating the query (Fernando et al. 2011). Overreliance on informal communication in distributed contexts may reportedly lead to incidents of miscommunication (Ramesh et al. 2012).

Since synchronous face-to-face conversations mediated by tools are often not feasible in a distributed context due to the temporal separation (e.g., Braithwaite and Joyce 2005), it is recommended to keep the amount of necessary interactions to a minimum. To achieve this, asynchronous as well as local meetings should be conducted (Hossain et al. 2009b). The number of necessary interactions may also be reduced by splitting the projects in parts that can be implemented independently (Szöke 2011). Furthermore, the degree of formality of the communication can be increased

by relying on detailed documentation such as architectural designs and plans (Ramesh et al. 2012; Szöke 2011). Nevertheless, even in distributed teams, it has been found that such formal documentation can be limited to what is absolutely needed (Sharp et al. 2012).

### 3.1.2 Team Size (Category 1b)

33 papers provide insight into the implications of the team size for communication. Since large teams tend to be distributed and smaller teams are rather co-located, findings in the team size category are also applicable to some degree to the previous section on team distribution. Specifically, communication is stated to be easier for smaller agile SD teams because collaboration becomes more and more complex as the team size grows (e.g., Melo et al. 2011). The major advantage reported for small teams is that informal face-to-face communication occurs naturally, which leads to the building of tacit knowledge (Kajko-Mattsson 2008; Nevo and Chengalur-Smith 2011).

Similarly to distributed environments, findings suggest that SD teams have to choose their communication mechanisms wisely in larger development situations as otherwise communication may be hindered (Pikkarainen et al. 2008). Reportedly, the adoption of agile SD practices also leads to improved communication in large-scale development projects (Petersen and Wohlin 2009, 2010). Informal communication may be problematic for large projects with many involved stakeholders and a lot of shared information (e.g., Cao et al. 2009; Fitzgerald et al. 2006; Pikkarainen et al. 2008). Synchronous and asynchronous communication media, tools, and channels are stated to be needed in larger projects in order to establish the multiplicity of social links between team members (e.g., Sarker et al. 2009) and to provide continual access to project information (Mishra et al. 2012). Some studies mention that more formal documentation for coordinating the large number of team members is needed; described media for coordinating large teams include wikis and group e-mails (Elshamy and Elssamadisy 2007; Hansson et al. 2006).

### 3.1.3 Project Domain (Category 1c)

Not much has been investigated with regard to the role of the project domain

for the communication of agile SD teams (four papers in total). Ramesh et al. (2010) investigate agile requirements engineering in the context of a mission-critical banking software. They find that frequent communication played an important role in the agile SD process. Although formal documentation was also used, the changing customer requirements could only be captured by frequent face-to-face communication. For security-critical applications, face-to-face communication with the customer is stated to be the most important source of requirements (Cao and Ramesh 2008). In terms of maintenance projects, first reported findings suggest that using only oral communication may become problematic later on during maintenance because implemented changes to the system are often not documented (Kajko-Mattsson 2008). Nawrocki et al. (2002) find that oral communication is susceptible to lapses of memory, especially when developing complex systems, since there may be problems recalling why certain alternatives were chosen. To avoid this problem, decisions should be documented.

## 3.2 Influence of Agile Methods and Practices on Communication Mechanisms

### 3.2.1 Findings on Extreme Programming (Category 2a)

95 papers include findings on the impact of Extreme Programming (XP) on communication. “XP is all about maximizing communication” (Murru et al. 2003, p. 39). Consequently, the most beneficial aspect of XP is reported to be more frequent communication within the team (Fruhling and de Vreede 2006; Layman et al. 2004; Ramesh et al. 2010). XP developers reportedly spend a quarter of their time on communication (Coman and Succi 2009). Furthermore, studies have found that agile practices of XP improve both formal and informal communication (e.g., Pikkarainen et al. 2008). Communication “at any level/phase of the XP project seems to be the backbone of all XP practices” (Juric 2000, p. 102). However, other findings suggest that the introduction of XP did not affect the way and how often the team communicated with the surrounding organization as well as with customer representatives (Svensson and Höst 2005).

**Table 2** Findings on XP practices

| XP practice                                     | # Papers | Impact on the communication mechanisms of SD teams                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Coding standards                                | 6        | Facilitates communication by giving all developers the same rules for developing code (Hazzan and Dubinsky 2003; Maruping et al. 2009b).                                                                                                                                                                                                                                                                                                                                         |
| Collective code ownership                       | 1        | Benefits communication indirectly by relying on timely communication between developers when sharing code (Xiaohu et al. 2004).                                                                                                                                                                                                                                                                                                                                                  |
| Continuous integration                          | 3        | Enables rapid feedback on the quality of the code (Xiaohu et al. 2004).                                                                                                                                                                                                                                                                                                                                                                                                          |
| On-site customer                                | 14       | Facilitates communication between the team and the customer by acting as a communication channel (e.g., Dorairaj et al. 2012a; Svensson and Höst 2005). Also entails a substantial positive influence on communication quality (Wojciechowski et al. 2010).                                                                                                                                                                                                                      |
| Pair programming                                | 47       | Fosters communication skills and improves interactions between the programmers (Karlström and Runeson 2006; Melo et al. 2011; Vidgen and Wang 2009).                                                                                                                                                                                                                                                                                                                             |
| Planning game                                   | 15       | Facilitates communication among team members and with the customer by forcing them to communicate in regular meetings (e.g., Cao et al. 2009; Dorairaj et al. 2012a).                                                                                                                                                                                                                                                                                                            |
| Refactoring                                     | 3        | Important for communication among developers (Hazzan and Dubinsky 2003) because the result of those practices is simplicity which is needed in order to communicate efficiently (Conboy et al. 2007).                                                                                                                                                                                                                                                                            |
| Simple design                                   | 3        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Small releases                                  | 4        | Enables more regular communication in meetings (Pikkarainen et al. 2008) and provides rapid feedback mechanisms between the customer and developers (Xiaohu et al. 2004).                                                                                                                                                                                                                                                                                                        |
| Sustainable pace                                | –        | We could not identify any study that deals with the “sustainable pace” practice and its implications for communication.                                                                                                                                                                                                                                                                                                                                                          |
| System metaphor                                 | 8        | Serves as a communication platform that is stated to improve mutual understanding and communication quality (Beck 1999; Dall’Agnol et al. 2003). It establishes a common vocabulary among developers and customers (Sharp and Robinson 2004). Many metaphors depending on different views of the system and even different stakeholders are stated to further improve communication (Keeling and Velichansky 2011).                                                              |
| Unit testing (incl. automated acceptance tests) | 9        | Automated acceptance tests are also mentioned to be an effective communication medium for customers and developers in order to exchange business requirements, domain knowledge, and the implementation status (Melnik et al. 2006; Park and Maurer 2009). Contradicting results are presented by Haugset and Hanssen (2008) who find that communication with customers is not improved because they lack understanding of the system so that specifying tests proves difficult. |

The most frequently investigated XP practice in terms of communication is pair programming. In total, 47 papers provide findings on pair programming and communication. Although “the purpose of pairing is to produce code, the process of pairing is fundamentally about communication—both verbal and non-verbal” (Sharp and Robinson 2006, p. 6). Pair programming is characterized by constant communication in an ad-hoc fashion between developers (Sharp and Robinson 2004; Sharp et al. 2006). It reportedly fosters communication skills and improves interactions between the programmers (Karlström and Runeson 2006; Melo et al. 2011; Vidgen and Wang 2009). Most of the conversations between pairs has been observed to be on a very low level of abstraction, including frequent questions (Stapel et al. 2010). It is reported that developers overall spend one third of their time without computer interaction during pair program-

ming (Plonka et al. 2011). Zarb (2012) observed the following communication topics during pair programming: talking about previous work; continuous review of the expected goal; explaining; silent instances; discussion; unrelated conversation; jokes; switching of roles; high 5 (celebration); and distraction. To sum up, “every study of pair programming has shown that it is a communications-intensive process” (Begel and Nagappan 2008, p. 125).

Compared with pair programming, studies investigating other XP practices are less frequent (cf. Table 2).

### 3.2.2 Findings on Scrum (Category 2b)

Although research on Scrum is increasing, there are remarkably fewer studies that address communication in relation to Scrum compared to XP (in total 64). Previous studies highlight that Scrum reminds the team members to talk about

the project and forces them to work together closely, which entails higher communication frequency and higher communication quality (e.g., Paasivaara et al. 2008; Paasivaara et al. 2009).

Table 3 summarizes our findings concerning the individual Scrum practices.

### 3.3 Influence of Communication Mechanisms on SD Success (Category 3)

We found 100 papers that discuss the relationship between communication and success of agile SD. Communication is repeatedly mentioned to be one of the most important success factors of agile SD projects (e.g., Aaen 2008; Asnawi et al. 2011; Hoda et al. 2011; Koskela and Abrahamsson 2004; Mishra and Mishra 2009; Nevo and Chengalur-Smith 2011; Overhage et al. 2011; Sharp et al. 2009). More specifically, it has been found that the higher the communication frequency, the more productive the project (Braithwaite

**Table 3** Findings on Scrum practices

| Scrum practice              | # Papers | Impact on the communication mechanisms of SD teams                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Daily scrum                 | 42       | <ul style="list-style-type: none"> <li>■ Is stated to be the most important mechanism for communication in Scrum (Pikkarainen et al. 2008).</li> <li>■ Communication is reported to be the glue that links all the projects members during the daily scrums (Ramesh et al. 2010).</li> <li>■ Is reported to not only have positive effects on the communication frequency among the whole team because one-to-one communication after the meetings is also encouraged (Paasivaara et al. 2009).</li> <li>■ All team members should participate in the meeting in order to avoid miscommunication (Wijnands and Dijk 2007).</li> </ul> |
| Sprint planning             | 13       | <ul style="list-style-type: none"> <li>■ Are used to share information on features and requirements with customer representatives (Pikkarainen et al. 2008).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| Sprint review/retrospective | 13       | <ul style="list-style-type: none"> <li>■ Also facilitate informal communication among the developers (Pikkarainen et al. 2008).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Scrum-of-scrums             | 9        | <ul style="list-style-type: none"> <li>■ Encourages communication between several teams (Paasivaara and Lassenius 2010).</li> <li>■ May create miscommunication and misunderstanding because of the reliance on a single person that represents the team (Hossain et al. 2009c).</li> </ul>                                                                                                                                                                                                                                                                                                                                           |

and Joyce 2005). Observed positive effects of improved communication on SD success in an agile SD environment include higher software quality (Hanakawa 2010), decreasing numbers of false features (Back et al. 2004) and less defects (Li et al. 2010), as well as shorter time and effort requirements (Molokken-Ostfold and Furulund 2007). Previous findings from case studies indicate that the effect of communication on team performance is mediated by factors such as trust (Paasivaara et al. 2008), mutual understanding (Holmström et al. 2006), and team identification (Piri and Niinimäki 2011).

Team members in SD projects using XP also supposedly understand the work better and solve tasks more quickly than teams which follow traditional development methods (e.g., Karlström and Runeson 2006; Layman et al. 2004). Developers have been found to benefit from intensified communication by less incidents of misunderstanding, increased efficiency of work, and better learning (Melnik and Maurer 2004; Vidgen and Wang 2009). Apart from communication among developers, strong customer communication is mentioned as a

key element of agile SD methods (e.g., Baskerville et al. 2002). Agile SD methods are reported to allow customers to communicate more precise and realistic requirements (Mann and Maurer 2005; Ramesh et al. 2010).

Informal face-to-face communication as the primary device for knowledge sharing has proven most effective, and it is suggested to utilize it as often as possible because it involves little filtering and distortion (Korkala et al. 2006; Mishra et al. 2012). Developers are also reported to strongly prefer face-to-face communication because immediate feedback is possible and misinterpretations can be easily avoided (LaToza et al. 2006). Specifically, it has been observed that less mistakes are made and less rework has to be done when important information is continuously shared through informal communication (Petersen and Wohlin 2009).

Especially in distributed environments, efficient communication is supposedly the key to ensuring high productivity and to fully profit from outsourcing benefits (e.g., Estler et al. 2012). A significant correlation has been found between distributed agile SD and communication

convergence, which in turn positively influences virtual team performance (Nevo and Chengalur-Smith 2011).

Despite this central role and positive reports, other studies find that communication is not significantly or even negatively related to the success of agile SD projects (Abbas et al. 2010; Misra et al. 2009). Communication is mentioned to be challenging when team members are engaged in multiple projects and the plan changes frequently, as well as when clear responsibilities are missing (Pikkarainen et al. 2012). Such challenges emerge, for example, when the feature list is not prioritized or subject to rapid changes and if there is not enough time for the planning meetings (Pikkarainen et al. 2008). The shift from traditional to agile methods also reportedly implies communication challenges when stakeholders expect a written, formal document but the team is prepared to focus on informal face-to-face conversations (Pikkarainen et al. 2008, 2012). In consequence, inadequate communication media can hinder communication between the team and external parties, and over-communication between team members or with the customer may also become an inhibitor (Vidgen and Wang 2009). Moreover, insufficient informal communication in terms of volume and diversity is the cause of many project challenges, but at the same time, too much informal communication is challenging and distracting for many individuals (Adolph et al. 2012). The emphasis on face-to-face conversations may also lead to challenges for team members with weak communication skills (Conboy et al. 2011).

### 3.4 Summary of Findings

**Table 4** summarizes the main findings of our literature review.

**Table 5** provides a general classification of the relevant papers that address communication within agile SD with regard to the employed research designs and methods (categories adapted from Urbach et al. 2009). The results show that there is an imbalance in terms of the employed research methods towards qualitative case studies, relying heavily on interviews. Other, more quantitative approaches such as questionnaire-based surveys or experiments are not conducted to the same extent. The same is true for design-oriented approaches; for example, most papers examine, describe,

**Table 4** Main findings of the literature on agile SD as regards the role of communication

| Categories                                                  | # Papers | Main findings in the literature in terms of impacts on communication                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Category 1a:</i><br>Team distribution →<br>Communication | 155      | <ul style="list-style-type: none"> <li>■ Communication is easier for co-located teams.</li> <li>■ Distributed teams face numerous challenges related to communication (e.g., different time zones).</li> <li>■ Agile SD helps to overcome those challenges by forcing the teams to communicate.</li> <li>■ Different communication modes are necessary in a distributed setting (asynchronous and synchronous media, enabled by tools).</li> </ul>                                                                                                                                                                              |
| <i>Category 1b:</i><br>Team size → Communication            | 33       | <ul style="list-style-type: none"> <li>■ Informal communication is mainly suited for small teams.</li> <li>■ Larger teams have to use more formal communication (e.g., for coordination and to document knowledge).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                  |
| <i>Category 1c:</i><br>Project domain →<br>Communication    | 4        | <ul style="list-style-type: none"> <li>■ When developing mission-critical systems, requirements should be communicated face-to-face in order to avoid miscommunication.</li> <li>■ In maintenance projects, more formal documentation is needed to retain knowledge.</li> </ul>                                                                                                                                                                                                                                                                                                                                                 |
| <i>Category 2a:</i><br>XP → Communication                   | 95       | <ul style="list-style-type: none"> <li>■ XP practices facilitate informal communication, especially the pair programming practice.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| <i>Category 2b:</i><br>Scrum → Communication                | 64       | <ul style="list-style-type: none"> <li>■ Scrum practices facilitate informal communication, especially the daily scrum practice.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <i>Category 3:</i><br>Communication → SD success            | 100      | <ul style="list-style-type: none"> <li>■ Positive impact: informal communication is reported as one of the key success factors of agile SD, especially in a distributed environment.</li> <li>■ Negative impact: informal communication is also reported to be challenging and overwhelming in case of many stakeholders or for maintaining knowledge.</li> <li>■ Identified mediators for communication and SD success include: mutual understanding, trust, and team identification.</li> <li>■ Observed key benefits include: higher productivity, higher software quality, need of less time and a lower budget.</li> </ul> |

and explain the use of agile SD in practice and do not attempt to design and evaluate new agile SD methods.

In order to further detail the extent to which research has investigated our search categories and to derive research gaps, we created systematic maps following the guidelines of Peterson et al. (2008). **Figure 2** presents the systematic map that depicts to what extent communication has been investigated in relation to XP practices, including the employed research methods. We find that research so far has mainly focused on pair programming, while research on other XP practices with regard to communication is lacking.

In terms of the Scrum practices, previous research has mostly been qualitative, mainly focusing on the daily scrum practice (cf. **Fig. 3**). Insights on other Scrum practices in relation to communication are less frequently present in the literature.

As regards our search categories SD success, project domain, team size, and team location, we also find many exploratory, qualitative studies on the impact of team distribution and team size as well as on the effect of communication on SD success (cf. **Fig. 4**). However, re-

search on other input factors such as the project domain is lacking.

## 4 Discussion

“Communication and collaboration are at the heart of agile software development” (Karhatsu et al. 2010, p. 298). Despite such statements and although communication is mentioned repeatedly as one of the most fundamental aspects of agile SD projects, our review shows that our current state of knowledge on the precise role of communication in agile SD and its impact on SD success is limited because previous results are scattered, inconclusive, as well as contradictory. Although communication is identified as one of the most fundamental aspects of agile SD projects, other studies report no or even negative effects of communication on performance and SD success. Communication is no ‘silver bullet’ for successful agile SD because over-communication may become an inhibitor, and communication hurdles arise in distributed or extended environments with many stakeholder groups and development teams. To sum up, we could not find any study that has conclusively

and rigorously shown that agile practices lead to more or improved communication in SD teams, and that this indeed in turn leads to higher SD success.

In terms of the input factors that influence the communication mechanisms of the agile SD process, the literature reports extensively on challenges of distributed teams and strategies to overcome those challenges. The systematic maps show that empirical evidence on team size is also available, but findings on the implications of the project domain on communication is lacking. Future research may also investigate whether there are other critical input factors that affect the communication mechanisms of agile SD teams. We find that most evidence in the input-related categories is based on qualitative and exploratory studies, whereas confirmatory and explanatory studies are lacking.

We also find that the literature refers to a very broad understanding of the concept “communication”. Previous studies indicate that communication is improved by means of following agile SD methods, but there are hardly any studies that open up the “black box” of the communication process itself or focus on the social interaction and behavior of teams as part of the research. For example, we

**Table 5** General characteristics of research on agile SD and communication

|                                          | # Papers | Percentage |
|------------------------------------------|----------|------------|
| <i>Research approach<sup>a</sup></i>     |          |            |
| Qualitative                              | 173      | 51.95 %    |
| Quantitative                             | 70       | 21.02 %    |
| Conceptualization (incl. Design Science) | 63       | 18.92 %    |
| Experience Reports                       | 39       | 11.71 %    |
| <i>Research method</i>                   |          |            |
| Case Study                               | 128      | 38.44 %    |
| Survey                                   | 26       | 7.81 %     |
| Experiment                               | 21       | 6.31 %     |
| Literature Review                        | 20       | 6.01 %     |
| Ethnography                              | 14       | 4.20 %     |
| Grounded Theory                          | 12       | 3.60 %     |
| Focus Group                              | 5        | 1.50 %     |
| Action Research                          | 5        | 1.50 %     |
| Simulation                               | 3        | 0.90 %     |
| None                                     | 103      | 30.93 %    |
| <i>Data collection techniques</i>        |          |            |
| Interviews                               | 103      | 30.93 %    |
| Questionnaire                            | 42       | 12.61 %    |
| Log Files                                | 36       | 10.81 %    |
| Field Notes                              | 33       | 9.91 %     |
| Video Recordings                         | 6        | 1.80 %     |
| Audio Recordings                         | 3        | 0.90 %     |
| Unknown                                  | 19       | 5.71 %     |
| None                                     | 109      | 32.73 %    |

<sup>a</sup>Percentages do not necessarily add up to 100 % because papers can contribute several times in the same category, for example, a paper can employ a qualitative as well as quantitative research approach

simply do not know whether there really is a higher or lower intensity, frequency, or quality in communication of agile SD teams in comparison to traditional, plan-driven teams. Rigorous empirical evidence of the consequences of the allegedly changing communication paradigm of agile SD on the development outcome is also scarce because of the exploratory nature of most studies. Do agile SD teams only perform better in specific contexts (which might be closely linked to communication)? May cases where better performance is observed really be attributed to improved communication (instead of other factors and processes)? We have not found any study that explicitly tries to compare or contrast traditional and agile SD as regards communication (e.g., using a field experiment). The systematic maps also show that research on other Scrum and

XP practices than pair programming and daily scrums is lacking. Consequently, we call for the rigorous investigation of the research gaps summarized in **Table 6**.

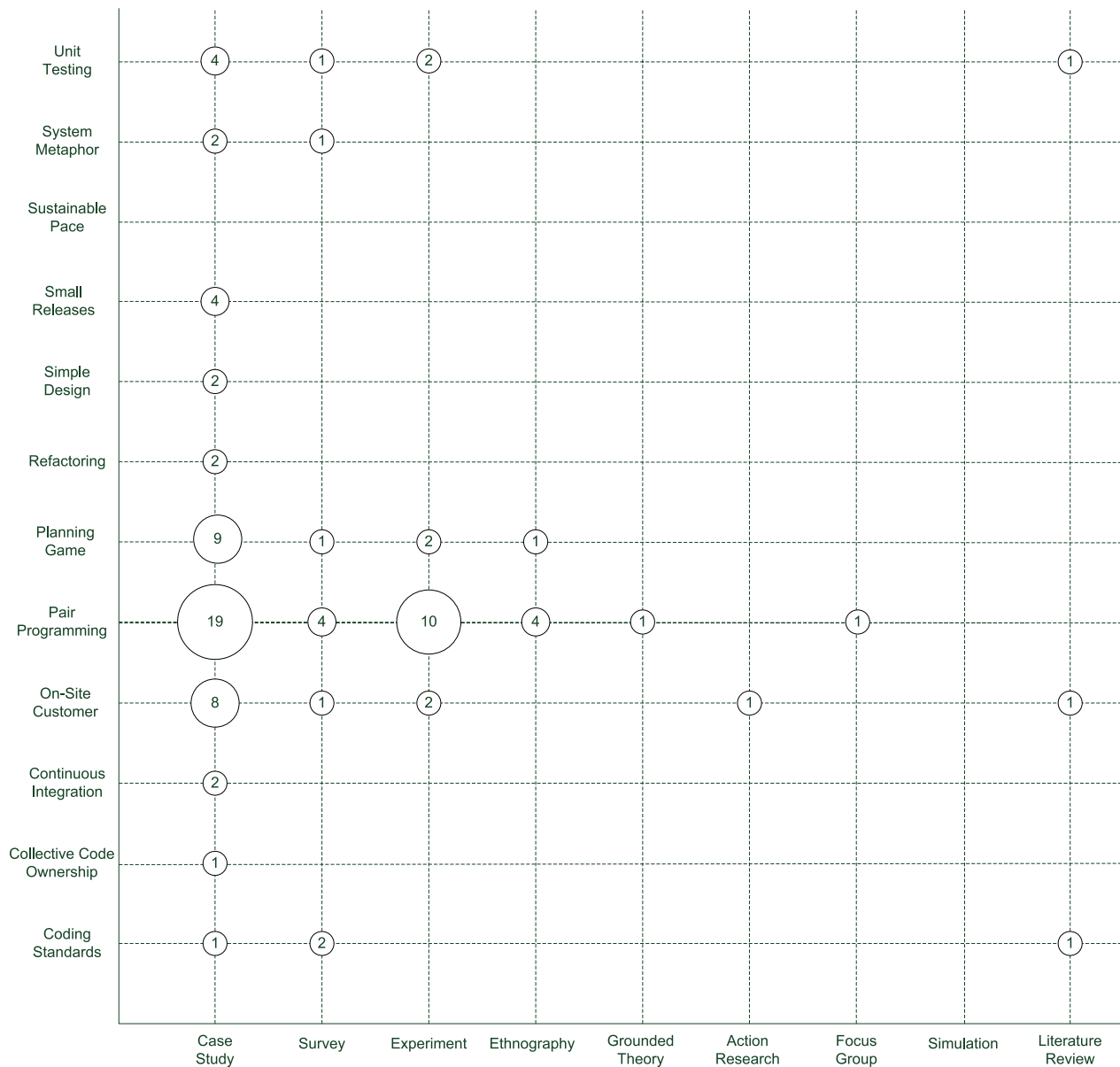
The identified gaps may be addressed using behavioral or design-oriented research approaches, but we explicitly call for confirmatory and explanatory assessments of the identified research gaps because our systematic maps show that first indications concerning many of these issues are already provided by practitioners and exploratory case studies. Gaining more insight into theoretical and empirical aspects of the research gaps depicted above will shed more light on our understanding of communication as a critical success factor of agile SD. Similarly, design-oriented studies combined with action-oriented research could investigate the effects of the use of different practices in distributed SD teams

in combination with novel tools aiming to support specific communication-related factors. All of this will lead to new knowledge on the underpinning theoretical foundations that constitute the success of agile SD.

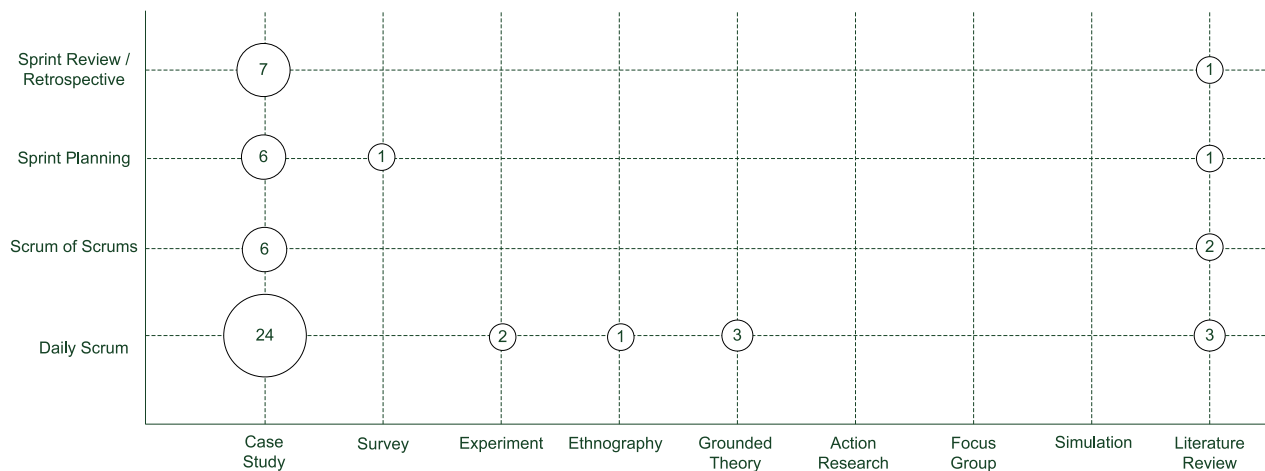
Despite our careful investigation, we may have missed some contributions due to our search procedure. We only searched for the keyword “communication”, and related keywords such as “collaboration” and “interaction” were not included because we believe that the additional search results would not have significantly improved the results. Instead of searching broadly for different keywords relating to communication, we decided to search deeply by scanning the full texts of relevant papers on agile SD for communication in depth. We are confident that we captured the essential findings of the literature because we expect that all papers that provide valuable insight on the subject agile SD and communication should at least mention communication in the title, abstract, or in the keywords. Furthermore, we may have missed findings on other agile methods than XP and Scrum since we did not include other agile methods in the search strings.

## 5 Conclusion

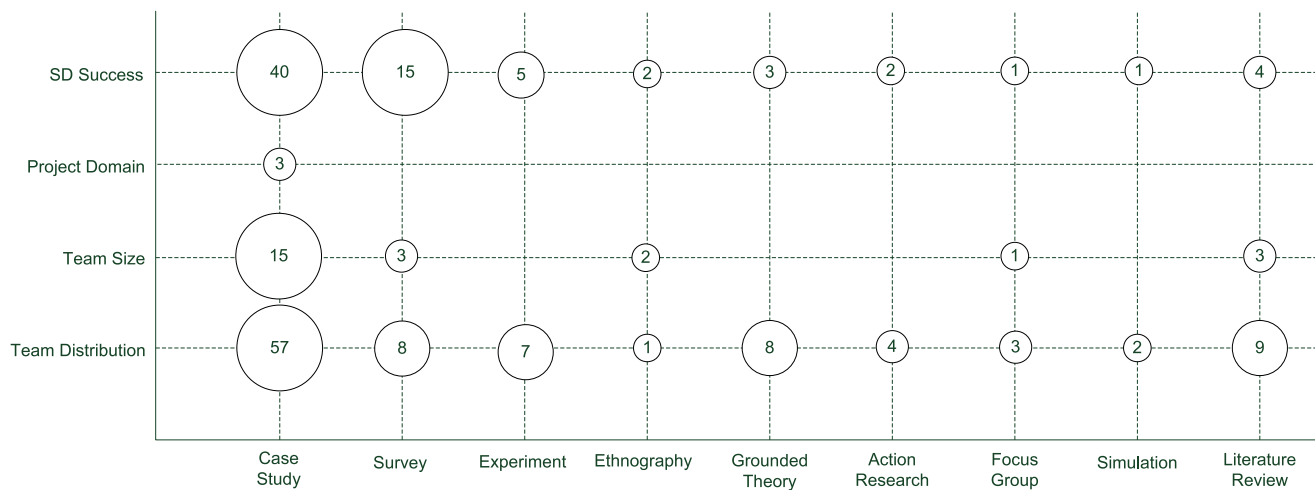
We aggregated existing knowledge on and insights into the role of communication within agile SD projects. Using the Unified Model of SD Success as a theoretical framework, our analysis highlights the value of communication for agile SD. However, it also shows that despite its importance, the communication process within agile SD is still not well understood. In consequence, we presented several research gaps that we feel need to be investigated in more detail. Investigating the role of communication along these lines may also help address one of the most pressing research issues: the missing “theoretical glue” (Conboy 2009, p. 330) of agile SD. For example, theories of communication, collaboration, cognition, and sense-making (e.g., Clark 1996; Garrod and Pickering 2009; Weick et al. 2005) could be used to build on the central role of communication in agile SD and could provide a starting point for studies trying to describe, explain, predict, and design the agile SD process.



**Fig. 2** Systematic map—research on XP practices and communication



**Fig. 3** Systematic map—research on Scrum practices and communication



**Fig. 4** Systematic Map—research on other categories and communication

**Table 6** Research gaps

| Category | Research gaps                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Input    | <p>What conditions surround and what factors on the environmental, organizational, group, and individual level impact on successful communication in agile SD?</p> <p>What is the impact of the project domain and SD context (factors on the environmental, organizational, group, and individual level) on the use and effect of communication mechanisms within agile SD teams?</p>                                                                          |
| Process  | <p>What is the impact of the Scrum practices on communication informality, frequency, and quality compared to traditional, non-agile SD teams? (Does the use of Scrum practices lead to more informal and better communication?)</p> <p>What is the impact of XP practices on communication informality, frequency, and quality compared to traditional, non-agile SD teams? (Does the use of XP practices lead to more informal and better communication?)</p> |
| Output   | <p>What are the implications of the changing communication paradigm of agile SD for the development outcome?</p> <p>Are agile SD teams more successful than traditional, non-agile SD teams due to improved communication? (Does more and better communication lead to better outcomes?)</p>                                                                                                                                                                    |

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## Abstract

Markus Hummel, Christoph Rosenkranz, Roland Holten

## The Role of Communication in Agile Systems Development

### An Analysis of the State of the Art

Agile systems development (SD) is described as a cooperative game of invention and communication, in which teams are characterized by relying heavily on informal communication mechanisms such as face-to-face conversations. Our current state of knowledge about the actual communication process in agile SD projects is fragmented because of the lack of an integrative lens across different studies. On the basis of the Unified Model of SD Success, we provide an integrated view on the role of communication in agile SD by conducting a structured, systematic literature review. Our data collection process resulted in 333 relevant papers on agile SD and communication. We find that despite its acknowledged importance, our knowledge on communication and agile SD is limited and research is scattered, inconclusive, as well as contradictory. In consequence, we present several research gaps that deserve future research attention. Specifically, we call for research that opens up the communication process and empirically validates the implications for communication frequency and quality triggered by agile SD.

**Keywords:** Agile systems development, Organizational communication, Literature review, State of the art, Software engineering

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